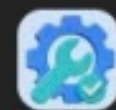




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FTC Troubleshooting Coach

Created by Teodora-Maria UDROIU (163150)

Our dead-wheel odometry robot is drifting 6 inches off-target to the left during our autonomous path.
Which software constants in our code should we change?



Diagnose a Problem

My [mechanism/subsystem] is



Full Troubleshooting Plan

We have an intermittent issue



Compare Two Design...

We're deciding between [option



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Before we jump to changing constants: what experience level are you (rookie, some experience, veteran, or mentor)?

A robot ending 6 inches left of the target does not automatically mean a software constant is wrong. With dead-wheel odometry, that symptom could come from:

- Incorrect track width (heading estimate error)
- Incorrect forward/lateral encoder scale factors (ticks-per-inch)
- A physical odometry wheel slipping or dragging
- An offset error in the odometry wheel positions
- A path follower tuning issue
- A starting pose mismatch

The highest-value next step is:

Run a straight-line autonomous test (for example, command 72 inches straight forward on a flat field) and measure:



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Run a straight-line autonomous test (for example, command 72 inches straight forward on a flat field) and measure:

1. How far the robot actually travels.
2. Whether it stays straight or drifts sideways.
3. What pose your odometry reports at the end.

Tell me:

- Which odometry library you're using (Road Runner, Pedro Pathing, custom, etc.).
- The commanded distance, actual distance, and final odometry-reported pose from that straight test.

With those measurements we can determine whether you should adjust encoder scale constants, track width, lateral multiplier, or something else, instead of guessing and tuning the wrong parameter.



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We just pushed it. The left encoder shows 4000 ticks, but the right encoder only shows 3200 ticks.

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That's a really useful measurement.

If you **pushed the robot straight by hand** and your odometry wheels are supposed to measure the same forward motion, then **4000 ticks on one side vs 3200 on the other is a 25% difference**, which is way too large to start by changing software constants.

Before touching track width or odometry multipliers, I'd treat this as a hardware/configuration issue.

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The single best next check is:

- Lift the robot so the odometry wheels can spin freely.
- Rotate the left and right odometry wheels exactly the same amount by hand (for example, 10 revolutions).
- Watch the encoder counts.

If one encoder still reports about 25% fewer ticks, then you likely have:

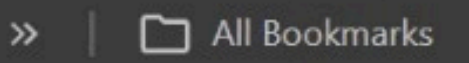
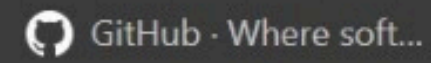
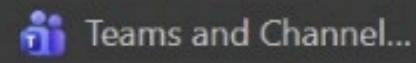
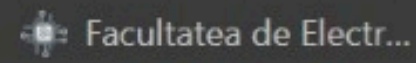
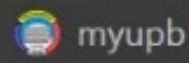
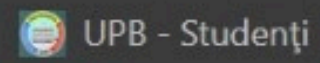
- Different ticks-per-revolution assumptions,
- An encoder configured incorrectly,
- A slipping coupler/gear,
- An odometry wheel that is not actually rotating 1:1 with the encoder shaft.

If the counts match when spun directly but not when pushing the robot, then look for:

- An odometry wheel losing contact with the floor ↓
- Different wheel diameters,

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- An odometry wheel that is not actually rotating 1:1 with the encoder shaft.

If the counts match when spun directly but not when pushing the robot, then look for:

- An odometry wheel losing contact with the floor,
- Different wheel diameters,
- Binding or drag,
- Slippage.

I would **not change any odometry calibration constants yet**. A 4000 vs 3200 tick difference from the same motion is evidence that the sensors are not measuring the same distance in the first place. Find out whether the mismatch comes from the encoder mechanism itself or from floor contact, then we can decide whether calibration is appropriate.



Give me a step-by-step encoder test

Help me check the encoder configuration

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Yeah, the tension spring on the right pod popped off so it wasn't pressing against the mat. We reattached it and now both read 4000 ticks when we push it straight.

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Nice, that's exactly the kind of evidence that helps isolate the cause.

Before:

- Left = 4000 ticks
- Right = 3200 ticks
- Right pod spring had come off

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After reattaching the spring:

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After reattaching the spring:

- Left = 4000 ticks
- Right = 4000 ticks

That makes the odometry wheel contact issue the leading explanation for the 6-inch left error.

When the right pod wasn't maintaining pressure on the mat, it was under-reporting motion, so the odometry solution was working from bad measurements.

At this point I would **not change any software constants yet.**

The next most useful test is to rerun the exact autonomous path with the repaired pod and compare:

- Final X error
- Final Y error
- Final heading error



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If the large left offset disappears or becomes much smaller, you've identified the root cause and avoided tuning around a hardware problem.

Only if there is still a consistent error after fixing the pod should you start looking at:

- Forward encoder scale calibration
- Lateral encoder scale calibration
- Track width calibration
- Path follower tuning

A good prevention step is to add "verify all odometry pods have spring tension and free wheel contact" to your pre-match checklist. Odometry failures like this can look like bad software and waste a lot of tuning time.

After you reran the autonomous, what was the new final error compared to the original 6-inch left miss?

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